Category Theory for Mathematical Physics — Exercise Sheet 4

Duality: cups, caps and quantum dimension

Variant: Questions only

Conventions as in Lecture 4: $\mathrm{ev}_X\colon X^*\otimes X\to 1$ is a cap and $\mathrm{coev}_X\colon 1\to X\otimes X^*$ a cup (Definition 4.1); a pivotal structure a gives the traces of Definition 4.6; in $\mathrm{Vect}_{\mathrm{fd}}$, a is the canonical $V\to V^{**}$. In Exercise 2, for a qubit, $|0\rangle, |1\rangle$ is the standard basis, X and Z are the Pauli matrices (elsewhere X is an object), and $|\Phi^{\pm}\rangle = (|00\rangle \pm |11\rangle)/\sqrt{2}$, $|\Psi^{\pm}\rangle = (|01\rangle \pm |10\rangle)/\sqrt{2}$. The letter j is a gauge number in Exercise 4 and an isomorphism $V\to V^*$ in Exercises 5 and 6, as in Definition 4.9; spins are written ℓ, so the lecture's $(-1)^{2j}a$ is $(-1)^{2\ell}a$ here.

Exercise 1 (Rotated boxes, a second dual and the cyclic trace). (i) In $\mathrm{Vect}_{\mathrm{fd}}$ let $f\colon X\to Y$, with bases (x_k) of X and (y_l) of Y, dual bases (x^k), (y^l), and $f(x_k)=\sum_l f_{lk}\,y_l$. Show that the rotated box f^* of Equation (4.2) has the transpose matrix of f.

(ii) Let $X=\mathbb{C}^2$ and $G=\begin{pmatrix}1&1\\0&1\end{pmatrix}$. Take $X'=\mathbb{C}^2$ (column vectors) with $\mathrm{ev}'(u\otimes x)=u^{T}Gx$. Find coev' making $(X',\mathrm{ev}',\mathrm{coev}')$ a left dual of X (check both snakes), and compute the isomorphism $\varphi\colon X^*\to X'$ of Theorem 4.2(a), with X^* written as row vectors.

(iii) Let $f=\begin{pmatrix}1&2&0\\0&1&3\end{pmatrix}\colon\mathbb{C}^3\to\mathbb{C}^2$ and $g=\begin{pmatrix}1&0\\0&1\\1&1\end{pmatrix}\colon\mathbb{C}^2\to\mathbb{C}^3$. Compute $\mathrm{tr}(g\circ f)$ and $\mathrm{tr}(f\circ g)$, and draw the three pictures of Equation (4.8) for this pair.

(iv) Cyclicity moves a box around the loop. Show that it does not allow other reorderings: find 2×2 matrices A,B,C with $\mathrm{tr}(ABC)\neq\mathrm{tr}(ACB)$. What, in the picture, stops C from jumping over B?

Exercise 2 (Bent wires and teleportation's four outcomes). (i) Draw the two maps of Equation (4.4), $\mathrm{Hom}(U\otimes V,W)\to\mathrm{Hom}(U,W\otimes V^*)$ and back, and show with one snake equation that bending a strand up and then down returns the morphism you started with. In $\mathrm{Vect}_{\mathrm{fd}}$ with $U=\mathbb{C}$, what does bending turn $f\colon\mathbb{C}^3\to\mathbb{C}^2$ of Exercise 1 into?

Alice holds $|\psi\rangle=\alpha|0\rangle+\beta|1\rangle$ and the first half of $|\Phi^+\rangle$; Bob holds the second half.

(ii) For each Bell state $|B\rangle\in\{|\Phi^+\rangle,|\Phi^-\rangle,|\Psi^+\rangle,|\Psi^-\rangle\}$ compute $(\langle B|\otimes\mathrm{id})(|\psi\rangle\otimes|\Phi^+\rangle)$, and write it as $\frac{1}{2}\sigma_B|\psi\rangle$.

(iii) Each Bell state is $(\mathrm{id}\otimes M_B)|\Phi^+\rangle$ for a 2×2 matrix M_B. Draw $\langle B|$ as the cap of Equation (4.5) with a box on its right leg, and explain σ_B by sliding that box along the snake to Bob's strand.

Exercise 3 (Super vector spaces). The objects of sVec are the finite-dimensional $\mathbb{Z}_2$-graded spaces $V=V_0\oplus V_1$; its morphisms are the linear maps preserving the grading. It has the tensor product with $(V\otimes W)_0=V_0\otimes W_0\oplus V_1\otimes W_1$, $(V\otimes W)_1=V_0\otimes W_1\oplus V_1\otimes W_0$, the unit $\mathbb{C}$ in degree 0, and the duals, ev and coev of $\mathrm{Vect}_{\mathrm{fd}}$, with $(V^*)_i=(V_i)^*$. Let P_V be $+1$ on V_0 and -1 on V_1.

(i) Show that $a'_V=a_V\circ P_V$ is a pivotal structure.

(ii) Show that its left trace is the *supertrace*, $\mathrm{str}(h)=\mathrm{tr}(h|_{V_0})-\mathrm{tr}(h|_{V_1})$, so that $d_V=\dim V_0-\dim V_1$, and that its right trace is also the supertrace.

(iii) Draw the picture proof of $d_{V\otimes W} = d_V d_W$ (Theorem 4.7): the loop on $V \otimes W$ as two nested loops. Then check $d_{V\otimes W} = d_V d_W$ directly from the grading of $V \otimes W$, and find a nonzero object of dimension 0.

Lecture 7 adds the braiding that makes this category "super"; Lecture 10 uses the pivotal structure a on it instead, with $d = +1$ on the odd line.

Exercise 4 (The semion snake). In Vec_G^ω on $\mathbb{Z}_2 = \{1, s\}$ with $\omega(s,s,s) = -1$, $s^* = s$ and $s \otimes s = 1$, so a cup $\mathrm{coev}_s = \lambda\ \mathrm{id}_1$ and a cap $\mathrm{ev}_s = \mu\ \mathrm{id}_1$ are numbers.

(i) Show that the first snake of Equation (4.1) is $\lambda\mu\,\omega(s,s,s)\ \mathrm{id}_s$, so both snakes are straight exactly when $\lambda\mu = -1$. What is the snake for the naive choice $\lambda = \mu = 1$?

(ii) A *gauge* changes the identification $s \otimes s = 1$ by a number $j \neq 0$: it is the monoidal functor that is the identity on objects with $J_{s,s} = j$ and the other J's and J_0 equal to 1. It sends (λ, μ) to $(\lambda/j, \mu j)$. Show that it is a monoidal functor, and that no gauge makes $\lambda = \mu = 1$ a solution. Is there a solution with $\lambda = \mu$?

Exercise 5 (Frobenius–Schur indicators of Q_8 and D_8). Let $Q_8 \subset \mathrm{SU}(2)$ be generated by $\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}$ and $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, and D_8, the symmetries of a square, by the rotation $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ and the reflection $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$. Each acts on $V = \mathbb{C}^2$ by these matrices, irreducibly.

(i) Show that both groups have order 8, and that the two characters on V take the value 2 once, -2 once and 0 six times.

(ii) Compute $\kappa_V = \frac{1}{8}\sum_g \chi_V(g^2)$ for each (the character formula of *Symmetries*).

(iii) Find a nonzero invariant bilinear form on V for each group, and check against Definition 4.9 that a nonzero symmetric invariant form gives $\kappa = +1$ and a nonzero antisymmetric one $\kappa = -1$. The two characters take the same values, with the same multiplicities; what group data, beyond these values, does κ see?

Exercise 6 ⋆. *The Penrose cap and the pivotal structure.* Let $V = \mathbb{C}^2$ be spin-$\frac{1}{2}$, with $\Omega(e_1, e_2) = 1 = -\Omega(e_2, e_1)$ as in Theorem 4.8, and $j\colon V \to V^*$, $j(v) = \Omega(\cdot, v)$.

(i) Show that $(\mathrm{id} \otimes j^{-1}) \circ \mathrm{coev}_V$ is the cup Ω and that $\mathrm{ev}_V \circ (j \otimes \mathrm{id})$ is the cap $-\Omega^\dagger$: the straight pair of Theorem 4.8 is (coev, ev) moved along j.

(ii) Show that the loop $-\Omega^\dagger \circ \Omega = -2$ is the quantum dimension $\mathrm{tr}_L(\mathrm{id}_V)$ for the map $a''_V\colon V \to V^{**}$ with $a''_V(v)(\varphi) = \Omega(j^{-1}\varphi, v)$, and that $a''_V = -a_V$.

(iii) On the finite-dimensional representations of SU(2) let $a'_W = a_W \circ \rho_W(-1)$. Show that a' is a pivotal structure, equal to a'' on spin-$\frac{1}{2}$; compute d_W for a' on spin ℓ, and the Frobenius–Schur indicator of spin-$\frac{1}{2}$ for a'.